

A Statistical Comparison of Sea Ice and Global Sea Level

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Abstract:

The accelerating pace of polar ice melt has raised significant concerns regarding its impact on global sea levels, particularly in coastal areas (Scott, 2023). This project aimed to comprehensively examine the relationship between polar ice melt and coastal sea level rise through a multidisciplinary approach. We used statistical analysis, climate models, and historical records, to quantify the contributions of polar ice loss to sea level rise using RStudio. By employing statistical techniques and modeling methodologies, we interpreted the spatial and temporal patterns of coastal sea level changes induced by polar ice melt. Through this process, we were able to prove a positive correlation between the rising sea levels and the accelerating polar ice melt.

Introduction and Hypothesis:

The Earth's polar regions, encompassing the Arctic and Antarctic, are undergoing unprecedented rates of ice loss due to climate change-induced warming (NASA, 2021). This accelerated polar ice melt has profound implications for global sea level rise, with coastal regions facing heightened risks of inundation, erosion, and flooding (Environmental Protection Agency, 2023).

We hypothesized that the accelerating polar ice melt, driven primarily by anthropogenic greenhouse gas emissions, is a significant contributor to the observed rise in coastal sea levels globally. We suggested that the magnitude and spatial distribution of coastal sea level rise are influenced by a multitude of factors, including the rate of ice loss from polar ice sheets and glaciers, oceanic circulation patterns, regional climate variability, and local geological factors. Furthermore, we anticipated that the impacts of accelerated polar ice melt on coastal sea levels would vary spatially and temporally, with some regions experiencing more pronounced sea level rise than others due to complex interactions between ice-ocean-atmosphere dynamics (Scott, 2023). Through a comprehensive investigation of these hypotheses, this project aimed to advance our understanding of the complex dynamic between polar ice melt and coastal sea level rise, thereby informing evidence-based decision-making for climate change adaptation and mitigation efforts in vulnerable coastal regions worldwide.

Methods:

Data collection:

We gathered satellite-derived data on polar ice extent and thickness from reputable sources from Kaggle. This included data that captured long-term trends in polar ice melt. Coastal sea level data was obtained from the same data source. These datasets provided information on variations in coastal sea levels over time and across different geographical regions.

Analysis:

First, our datasets were not fully aligned by date. So, to fix this, we sorted the data into subsets with the same range of years.

```
> sea_level_subset <- sealevel %>%  
+   filter(Year >= 1978 & Year <= 2013)  
> sea_ice_subset <- seaice %>%  
+   filter(Year...1 >= 1978 & Year...1 <= 2013)
```

However, from this, we ran into another issue. There were multiple data values per year, we wanted to merge the datasets, but we needed to maintain the order for each. To do this we assigned a number order to each row.

```
> sea_ice_subset <- sea_ice_subset %>%  
+   group_by(Year...1) %>%  
+   mutate(row_number_ice = row_number())  
> sea_level_subset <- sea_level_subset %>%  
+   group_by(Year) %>%  
+   mutate(row_number_level = row_number())
```

From this, we were able to merge the data subsets seamlessly.

```
> merged_data <- inner_join(sea_ice_subset, sea_level_subset, by =  
c("Year...1" = "Year", "row_number_ice" = "row_number_level"))
```

Great! We are finally able to do our analysis. We conducted a Pearson correlation test and got a coefficient of -0.48.

```
> correlation <- cor(merged_data$Extent, merged_data$GMSL_noGIA)  
> print(correlation)  
[1] -0.4836917
```

Which brings us to...

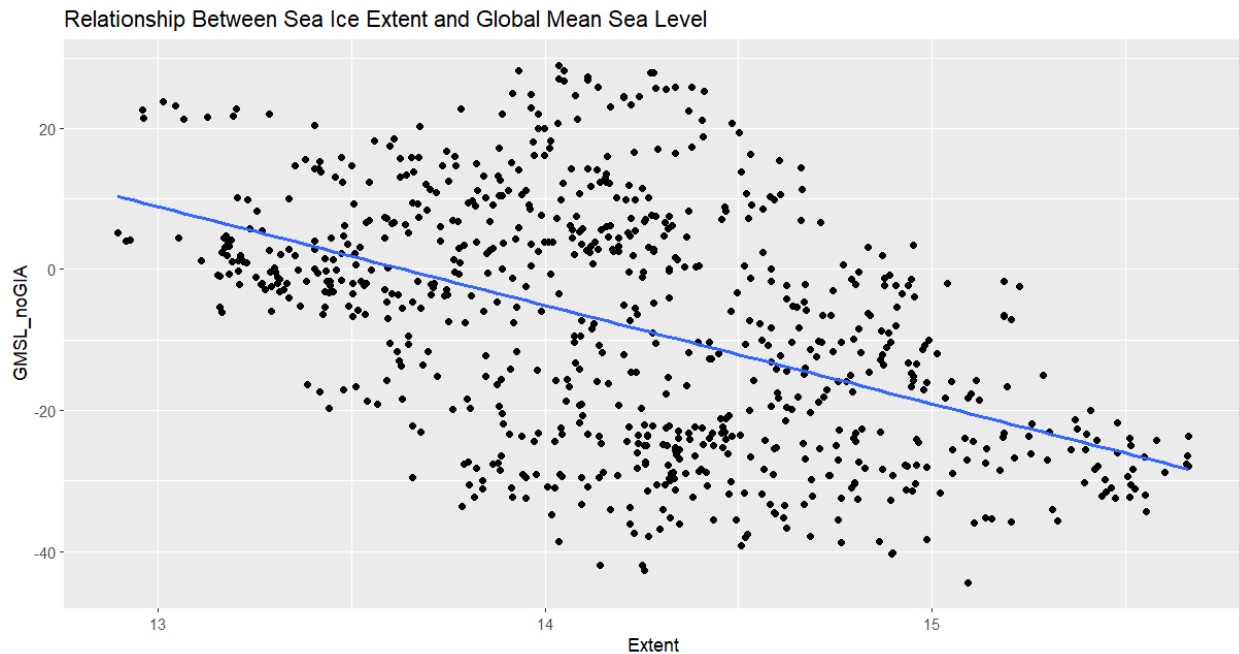
Results

From that correlation test, we were able to plot sea ice extent versus global mean sea level rise with a line of best fit.

```
> ggplot(data = merged_data, aes(x = Extent, y = GMSL_noGIA)) +  
+   geom_point() +  
+   geom_smooth(method = "lm", se = FALSE) +  
+   labs(title = "Relationship Between Sea Ice Extent and Global Mean Sea  
Level",
```

```
+      x = "Extent",  
+      y = "GMSL_noGIA")
```

Which gave us the following graph:



From a correlation coefficient of -0.48, we can determine that there is a moderate negative correlation between sea ice extent and global mean sea level. As sea ice extent increases, global mean sea level tends to decrease, and vice versa. To further test our results, we constructed a hypothesis test to obtain a p-value and a T-value:

```
> cor_test <- cor.test(merged_data$Extent, merged_data$GMSL_noGIA)  
> print(cor_test)
```

```
data: merged_data$Extent and merged_data$GMSL_noGIA  
t = -15.345, df = 771, p-value < 2.2e-16  
alternative hypothesis: true correlation is not equal to 0  
95 percent confidence interval:  
 -0.5359275 -0.4277668  
sample estimates:  
      cor  
-0.4836917
```

The p-value associated with the correlation coefficient is less than $2.2e-16$, which is extremely small. This indicates strong evidence against the null hypothesis that there is no correlation between sea ice extent and global mean sea level. The t-value associated with the correlation coefficient is -15.345, indicating the strength of the correlation relative to the data's variability. The larger the absolute value of the t-value, the stronger the evidence against the null hypothesis.

Conclusion

Based on these results, we concluded that there is a statistically significant negative correlation between sea ice extent and global mean sea level in the dataset. This means that as sea ice extent decreases, global mean sea level tends to increase, and vice versa. We can reject our null hypothesis.

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[melt#:~:text=Together%2C%20the%20Antarctic%20and%20Greenland,to%20affect%20population%20centers%20worldwide](https://nsidc.org/learn/ask-scientist/where-will-sea-level-rise-most-ice-sheet-melt#:~:text=Together%2C%20the%20Antarctic%20and%20Greenland,to%20affect%20population%20centers%20worldwide)

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